Math 526, S06 **Name:**
Quiz 9 **SSN: xxx-xx-**

Instructions: **There are totally 2 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.**

Problem 1. (12 points) Consider the system $\mathbf{A}\mathbf{x} = \mathbf{b}$ where

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 0 \\ 1 & -4 & 2 \\ 2 & -1 & 4 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} -1 \\ -1 \\ 5 \end{bmatrix}$$

a) Write down the corresponding $\mathbf{S}$, $\mathbf{T}$ and the detailed iterative formula for the Gauss-Seidel method. **Solution:** The Jacobi method:

$$\mathbf{S}\mathbf{x}^{new} = \mathbf{b} + \mathbf{T}\mathbf{x}^{old}$$

$$\mathbf{S} = \begin{bmatrix} 2 & 0 & 0 \\ 1 & -4 & 0 \\ 2 & -1 & 4 \end{bmatrix}, \qquad \mathbf{T} = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

i.e.,

$$\begin{aligned} x_1^{new} &= (-1 - x_2^{old})/2 \\ x_2^{new} &= (-1 - x_1^{new} - 2x_3^{old})/(-4) \\ x_3^{new} &= (5 - 2x_1^{new} + x_2^{new})/4 \end{aligned}$$

b) Starting with the initial guess $\mathbf{x}_0 = \mathbf{0}$, compute $\mathbf{x}_1$ and $\mathbf{x}_2$ using the Gauss-Seidel method.

$$\mathbf{x}_1 = \begin{bmatrix} (-1-0)/2 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1/2 \\ (-1-(-1/2)-2\cdot 0)/(-4) \\ x_3 \end{bmatrix}$$

$$= \begin{bmatrix} -1/2 \\ 1/8 \\ (5-2\cdot(-1/2)+1/8)/4 \end{bmatrix} = \begin{bmatrix} -1/2 \\ 1/8 \\ 49/32 \end{bmatrix}$$

$$\mathbf{x}_2 = \begin{bmatrix} (-1-1/8)/2 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -9/16 \\ (-1-(-9/16)+2\cdot 49/32)/(-4) \\ x_3 \end{bmatrix}$$

$$= \begin{bmatrix} -9/16 \\ 7/8 \\ (5-2\cdot(-9/16)+7/8)/4 \end{bmatrix} = \begin{bmatrix} -9/16 \\ 7/8 \\ 7/4 \end{bmatrix}$$

Problem 2. (8 points) Let $\mathbf{A} = \begin{bmatrix} 2 & 2 & 3 \\ 1 & -1 & 1 \\ 3 & 2 & 1 \end{bmatrix}$. Compute $det(\mathbf{A})$. Is $\mathbf{A}$ singular?

Solution:

$$\begin{aligned} det(\mathbf{A}) &= 2 \cdot det \begin{bmatrix} -1 & 1 \\ 2 & 1 \end{bmatrix} - 2 \cdot det \begin{bmatrix} 1 & 1 \\ 3 & 1 \end{bmatrix} + 3 \cdot det \begin{bmatrix} 1 & -1 \\ 3 & 2 \end{bmatrix} \\ &= 2 \cdot (-3) - 2 \cdot (-2) + 3 \cdot 5 \\ &= 13 \end{aligned}$$

So $\mathbf{A}$ is nonsingular.